

Deep Q-Networks and Double DQN for Atari Breakout

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Course: Introduction to Deep Reinforcement
Learning

Degree Program: M.Sc. Computer Science

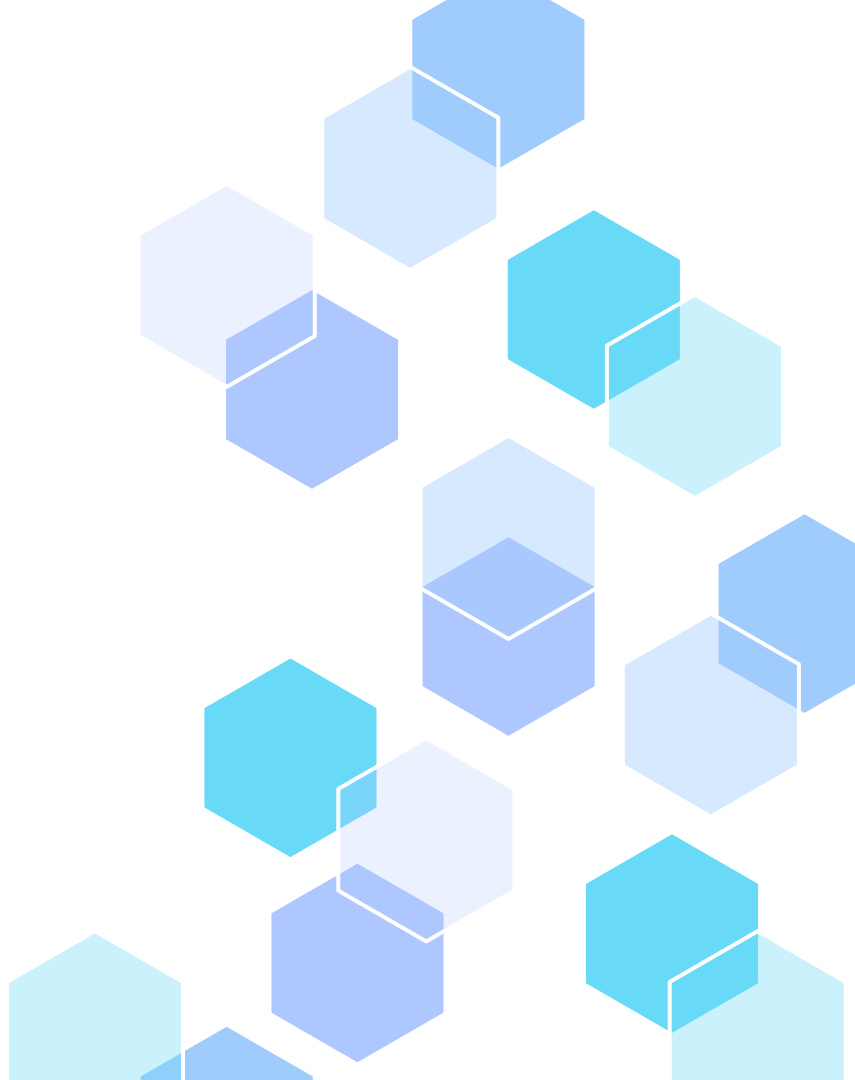
Kiel University

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Project Objectives



Goal of the Project

- Implement DQN and Double DQN
- Train agents on Atari Breakout
- Compare learning performance and stability
- Investigate the impact of overestimation bias



Brief DQN Recap

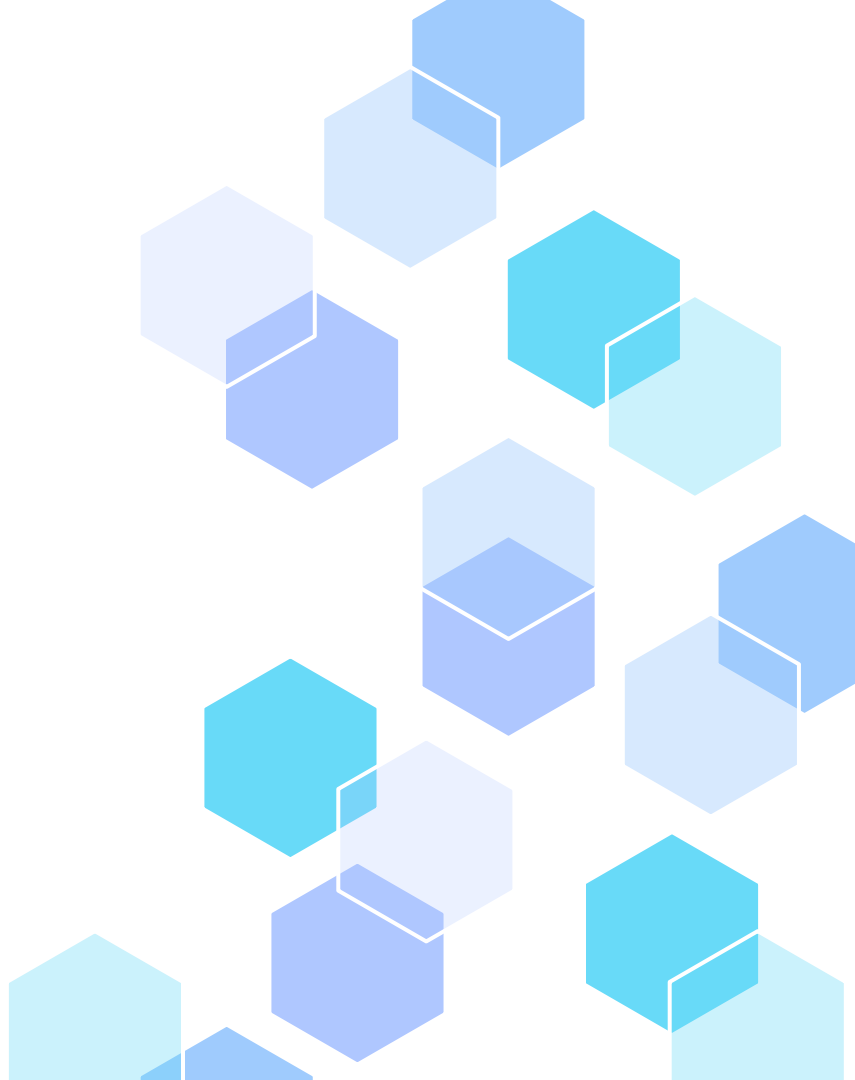
- DQN approximates Q-values using a neural network.
- Uses:
 - Experience Replay
 - Target Network
- Successfully learns from image observations.

State $\rightarrow$ CNN $\rightarrow$ Q-values $\rightarrow$ Action



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Motivation for Double DQN



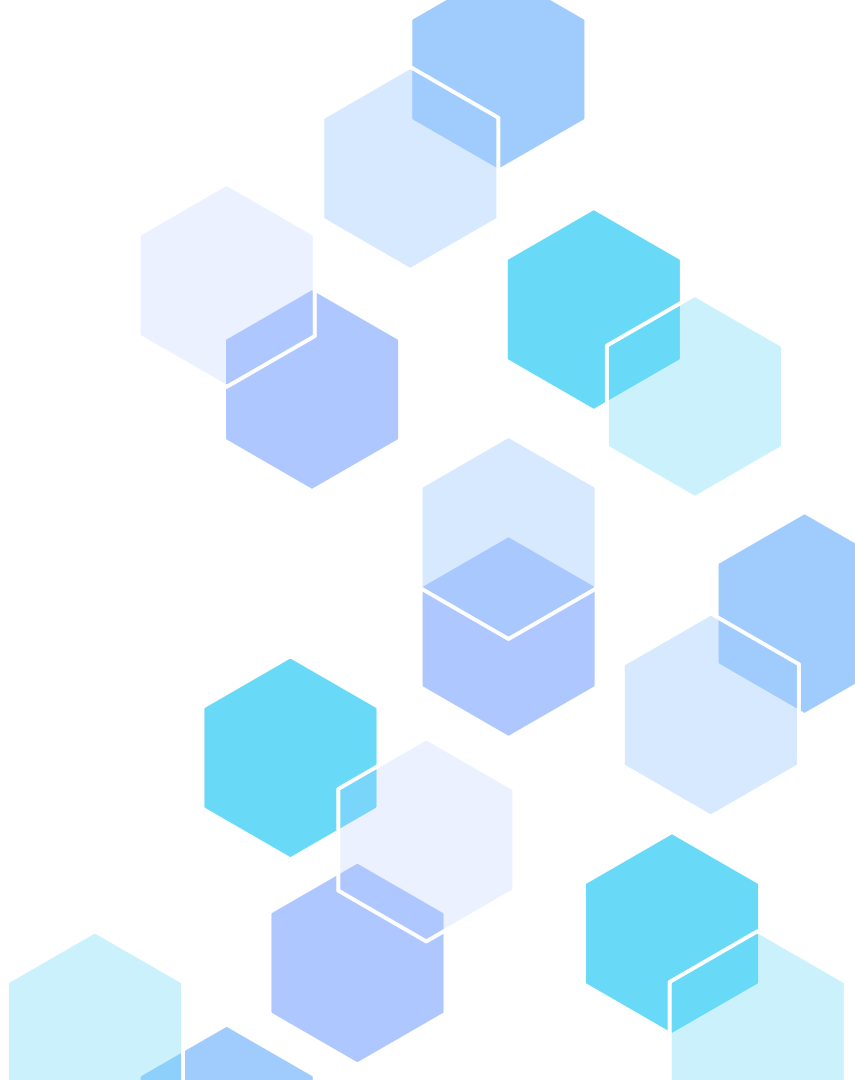
Problem: Overestimation Bias

- DQN uses the same network to:
 - Experience Replay
 - Target Network
- Can lead to overly optimistic Q-values.
- May result in unstable learning.



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Double DQN



Key Idea

- Online Network selects action.
- Target Network evaluates action.



Target Equations

DQN

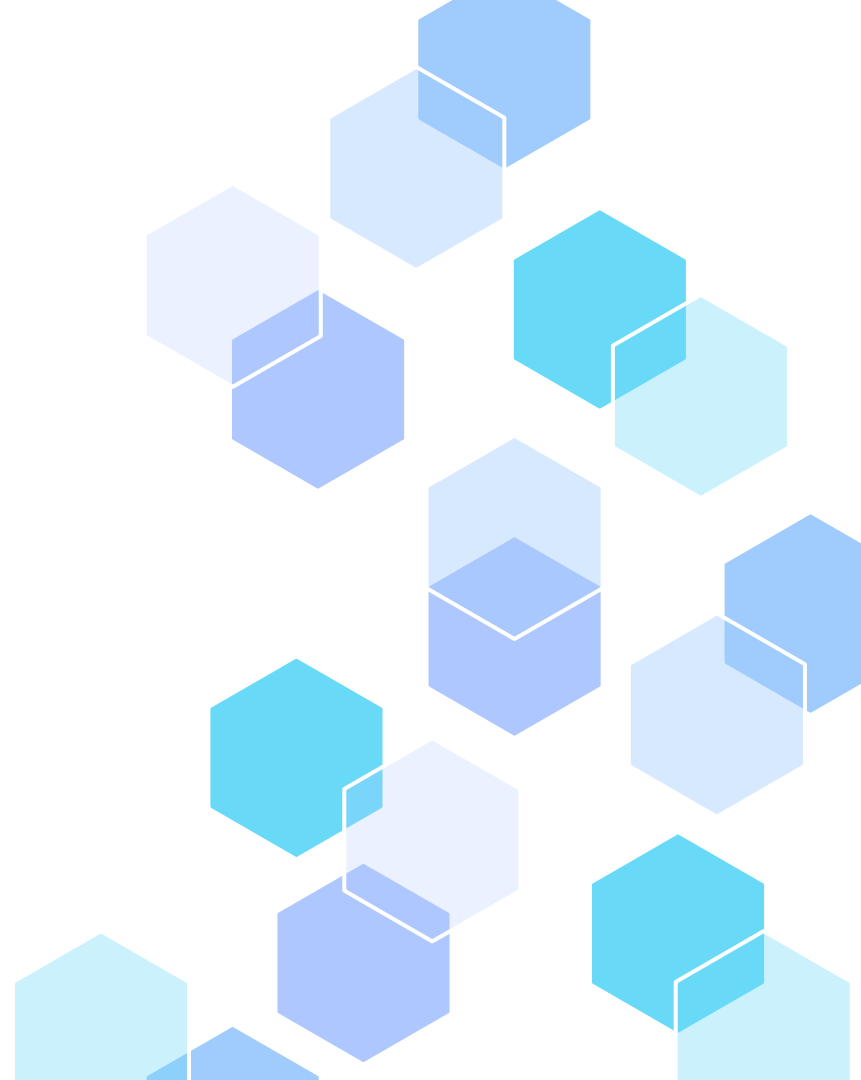
$$y = r + \gamma \max_a Q_{target}(s', a)$$

Double DQN

$$y = r + \gamma Q_{target}(s', \arg \max_a Q_{online}(s', a))$$

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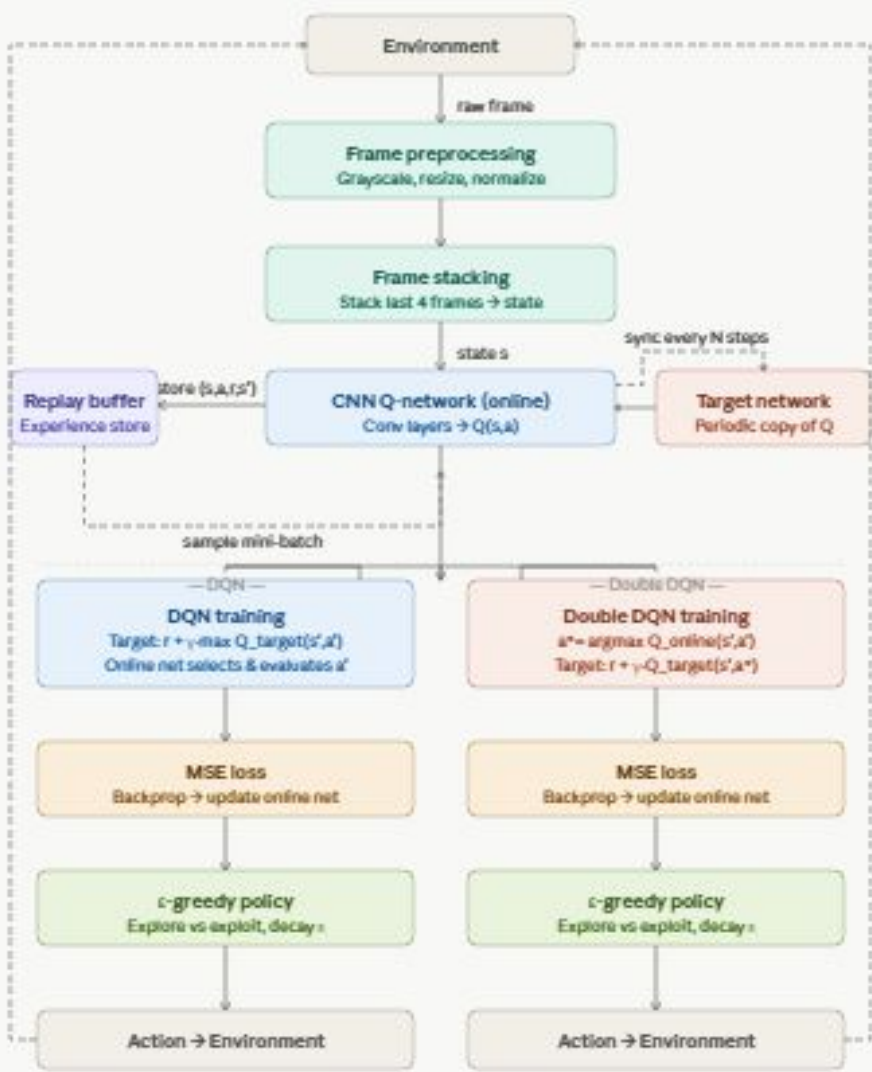
Experimental Setup



Implementation Overview

Components:

- Frame preprocessing
- Frame stacking
- Replay buffer
- CNN-based Q-network
- Target network
- DQN and Double DQN training



Model Architecture

Component	Details
Q-Network	CNN + Fully Connected Layers
Activation	ReLU
Output	Q-Values for each action
Target Network	Periodically updated copy of Q-Network

Training Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Learning Rate	1e-4
Discount Factor (γ)	0.99
Batch Size	32
Replay Buffer Size	100K steps
Training Frequency	4
Target Update Frequency	1000

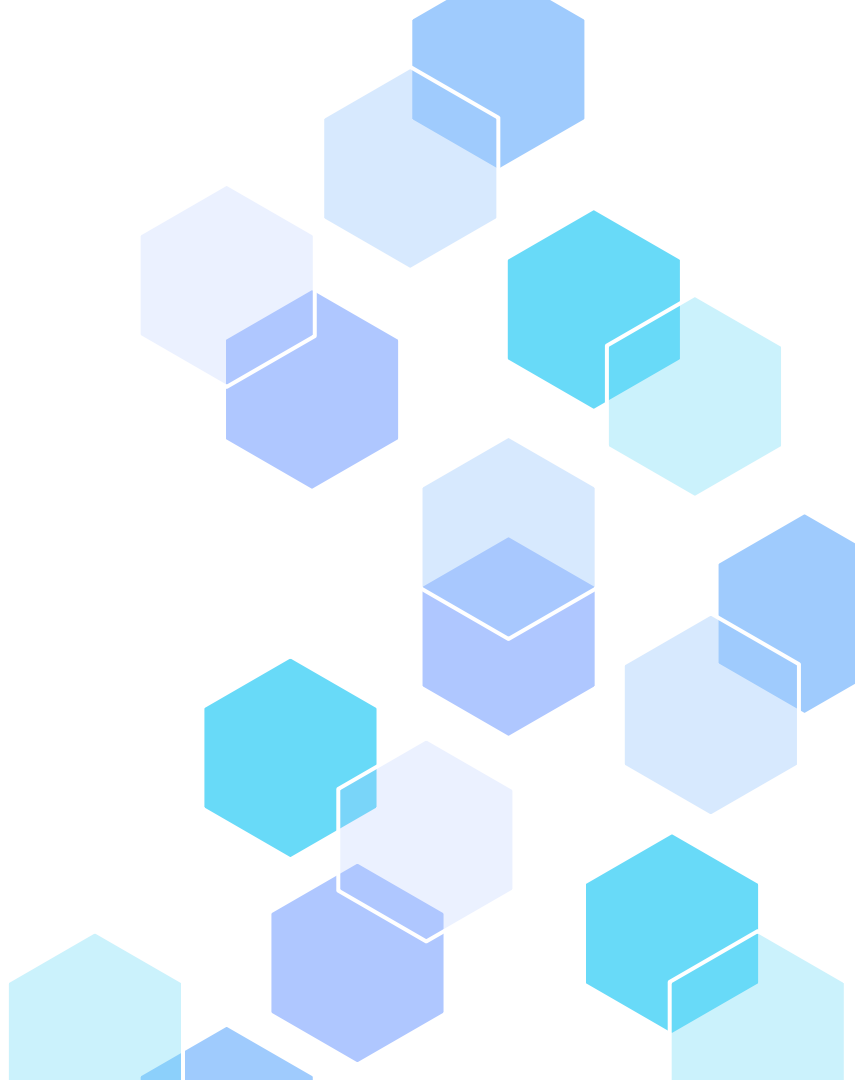
Exploration Strategy

Component	Details
Policy	ϵ - Greedy
Initial ϵ	1.0
Final ϵ	0.1
Decay Strategy	Linear Decay

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Results after first 100K Steps

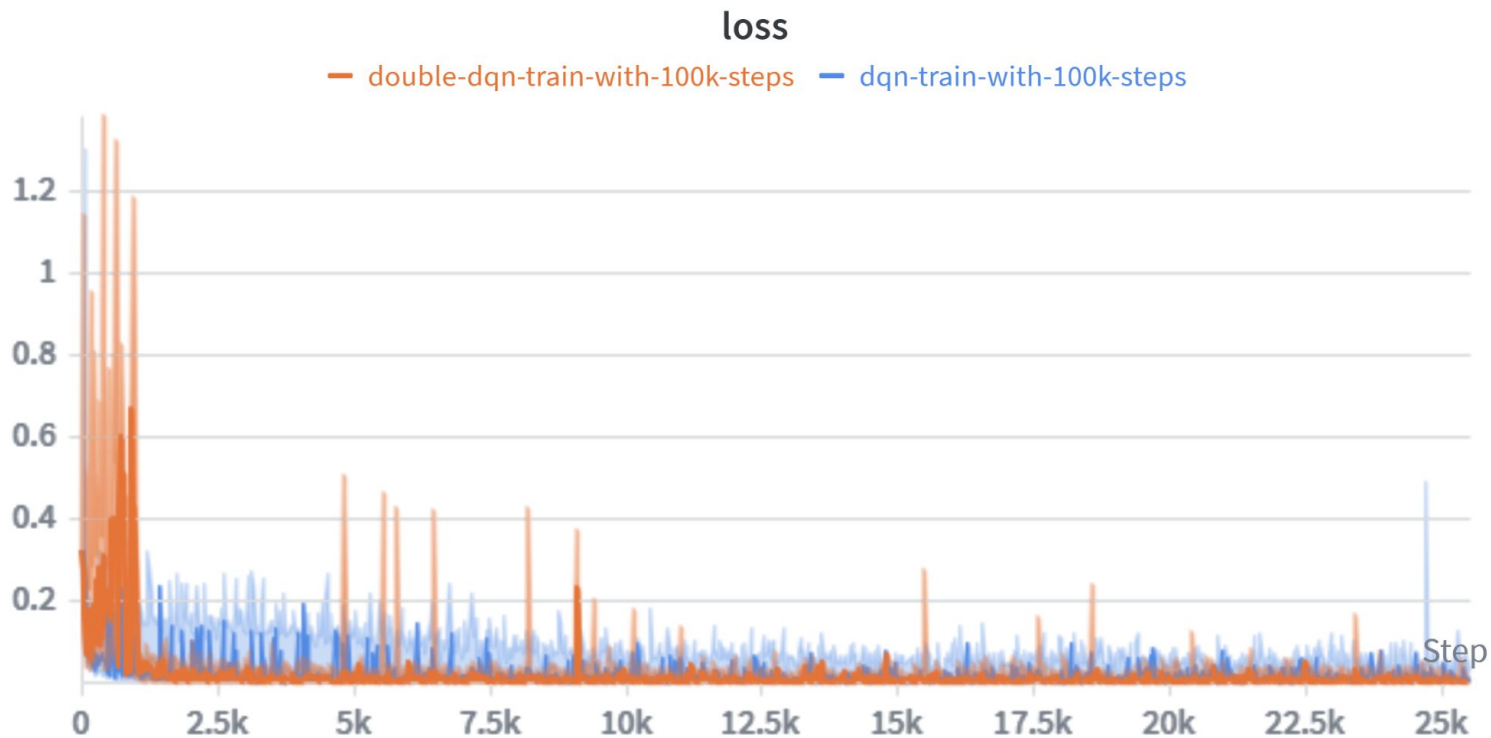
Linear decay of epsilon from 1.0 to 0.1



Reward Curves



Loss Curves



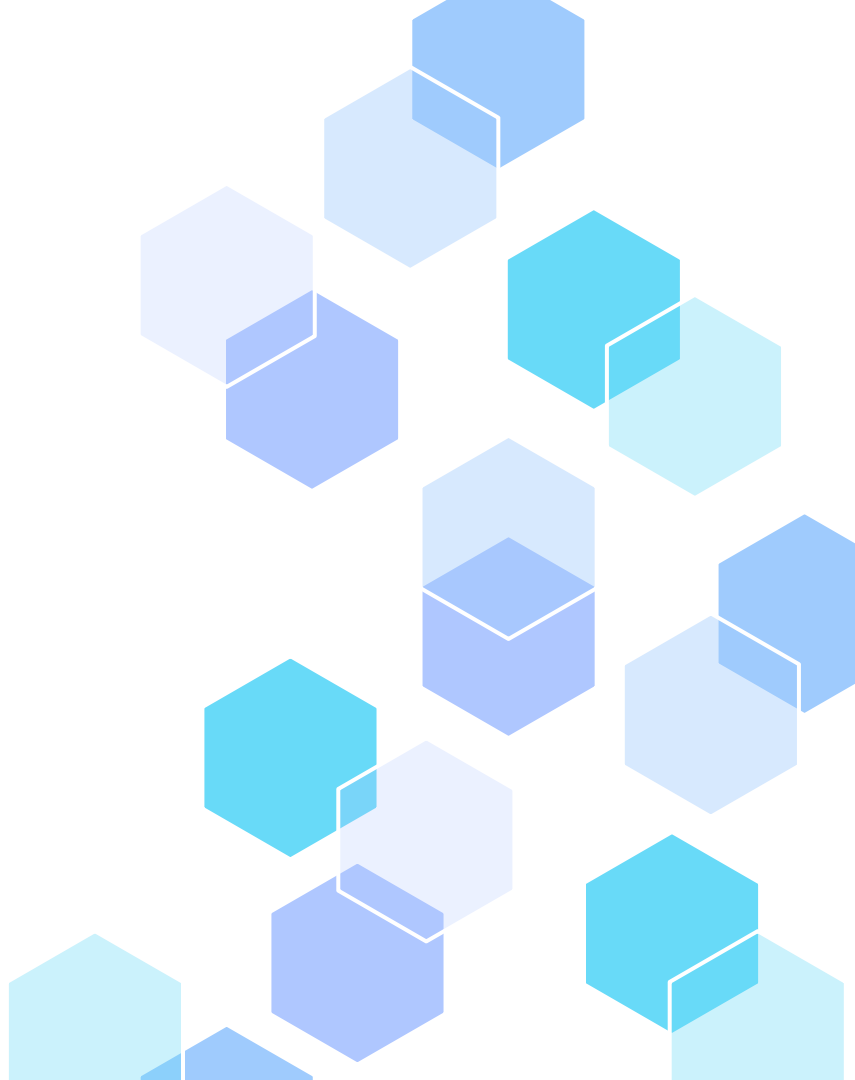
Results after 100K Steps

Metric	DQN	Double-DQN
Avg. Reward	2.48	2.28
Max. Reward	8	8
Std Dev	1.69	1.28
Loss	0.010	0.003

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Extended Training (200K Total Steps)

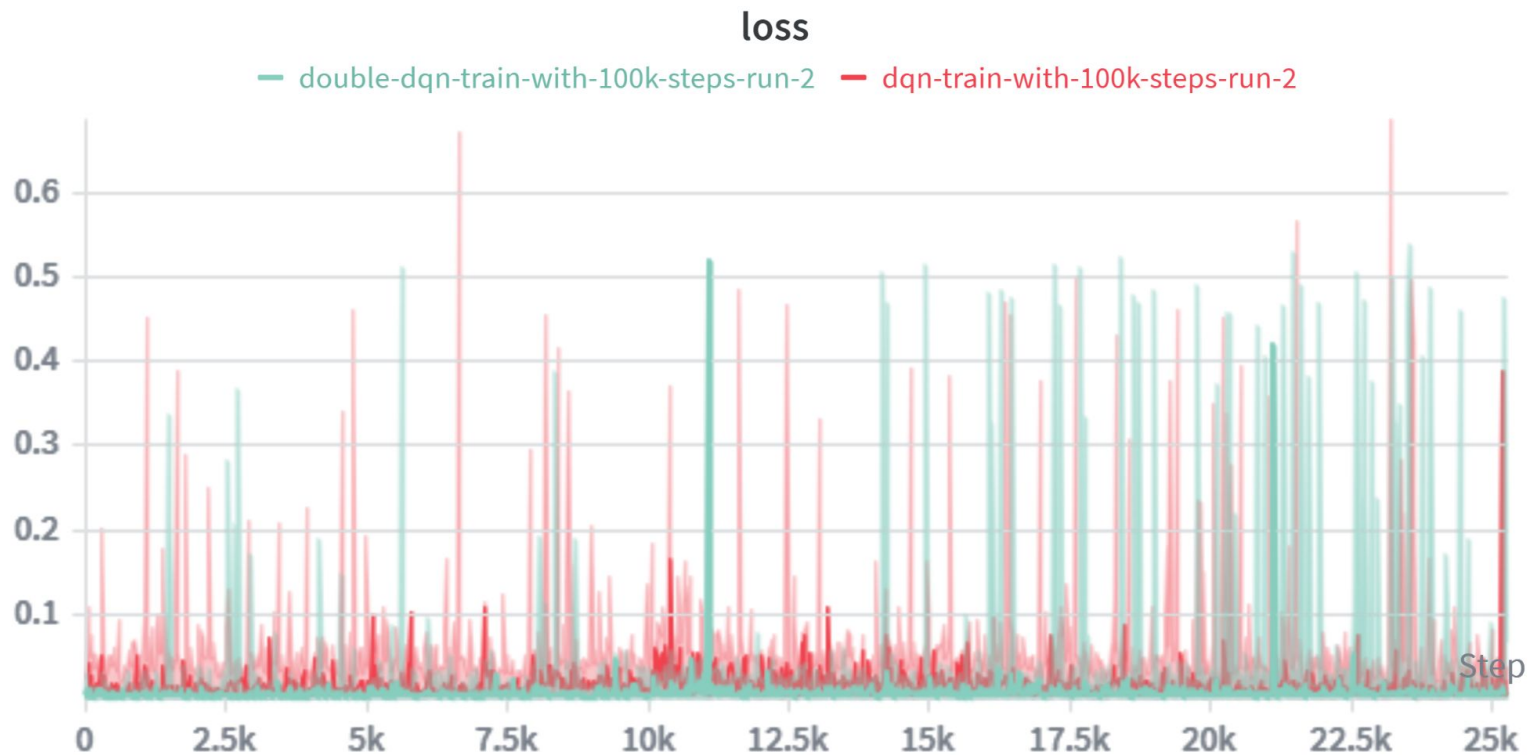
Constant epsilon of 0.1
Step = 100K to 200K



Reward Curves



Loss Curves



Results after 200K Steps

Metric	DQN	Double-DQN
Avg. Reward	7.88	11.08
Max. Reward	18	26
Std Dev	3.04	4.05
Loss	0.008	0.002

Improvement

- DQN: Avg Reward = 7.88
- Double DQN: Avg Reward = 11.08
- Double DQN outperformed DQN by ~41%



Analysis and Discussion

- DQN and Double DQN similar initially.
- Double DQN benefits become visible with longer training.
- Double DQN has lower losses throughout training.
- Double DQN has higher final rewards.



Conclusion

- Successfully implemented DQN and Double DQN.
- Double DQN reduced overestimation bias.
- Achieved higher long-term performance in Breakout.
- Results support the theoretical advantages of Double DQN.



Future Work

- Prioritized Replay
- Dueling DQN
- Longer training runs
- Multiple random seeds

